



A parameterized physics-informed machine learning approach for solving heat and mass transfer equations in the drying process

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ABSTRACT

This paper introduces a novel application for surrogate modeling in the context of coupled heat and mass transfer during drying using physics-informed neural networks (PINNs). The Luikov model is implemented within the PINNs model to simulate heat and mass transfer dynamics. The proposed method enables the integration of crucial parameters like the thermal Biot number (Bi_t) and mass transfer Biot number (Bi_m) in the model's architecture, facilitating a versatile framework to address a wide range of operating conditions. The range of variation for the Bi_t lies between 0 and 1, while the range of the Bi_m is between 0 and 10. The results predicted by PINNs model has been validated against reference values in literature obtained via Finite Element Method (FEM), analytical solution, and experimental data to verify the model's accuracy. Sensitivity analysis revealed that temperature sensitivity to Bi_t peaked at an optimal range, while moisture sensitivity remained negligible. The model's capability to predict outcomes across varying Bi_m values was confirmed, showing high accuracy in capturing the dynamics of the drying process. Sample refinement enhanced prediction accuracy, especially in the critical range of Bi_m values.

1. Introduction

Heat and mass transfer processes are fundamental in numerous aspects of life science engineering, encompassing critical phenomena such as thermal regulation in biological systems [1], drug formulation [2], food processing [3], cooling [4], wood thermal modification [5,6], and tissue engineering [7]. Accurate and efficient modeling of these processes is paramount for optimizing design, enhancing product quality and energy efficiency, and ensuring safety in various industrial applications. Throughout the decades, conventional numerical approaches have stood as the cornerstone of scientific and engineering simulations, aiding our understanding of complex physical processes and systems [8].

In this respect, many researchers have dedicated their efforts to tackling the subtle challenges associated with heat and mass transfer. For example, Mondal and Datta [9] performed a CFD simulation of the crustless bread-baking process using the finite element method by solving the unsteady heat and mass transfer equations. The study of Luikov Eqs. [10] has captivated the attention of scientists and researchers in the realm of heat and mass transfer due to their distinctive capability to model the intricate dynamics of simultaneous heat and

mass transfer in porous media. Chang and Weng [11] introduced an analytical method utilizing Laplace transformation to efficiently solve coupled heat and moisture transfer equations in porous materials. He et al. [12] focused on enhancing the quality of wood, a renewable and natural material predominantly used in building applications by establishing and solving a heat and mass transfer model using COMSOL Multiphysics software. The investigation established key relationships

between temperature and moisture content, medium temperatures during thermal treatment, and treatment duration.

However, conventional methods suffer significant costs when addressing optimization problems that demand repetitive simulations for each design parameter. Moreover, they encounter challenges in solving inverse problems [13] and handling ill-posed boundary conditions [14]. In the contemporary landscape, the fusion of artificial intelligence (AI) and deep learning techniques with established scientific methodologies signals a transformative era in computational modeling and simulation. AI-driven methods, particularly deep learning, have emerged as powerful tools capable of autonomously learning complex process dynamics directly from data, offering a promising avenue for enhanced efficiency and accuracy in simulation tasks [15,16]. Recent studies have demonstrated the potential of AutoML in various engineering applications. For instance, Krzywanski et al. [17]

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Nomenclature			
x	Length (m)	$N(X; \theta)$	Neural Network with input X and parameter θ
t	Time (s)	Lu	Luikov number
T	Temperature (K)	Bi_m	Mass transfer Biot number
M	Moisture content $\left(\frac{kg}{kg}\right)$	Bi_t	Thermal Biot number
C_p	Specific heat capacity $\left(\frac{J}{kg \cdot K}\right)$	L_v	Latent heat of moisture evaporation
D	Effective diffusion	Pn	Posnov number
X	Non-dimensional length	Ko	Kossovich number
ρ	Density $\left(\frac{kg}{m^3}\right)$	r	Radius (m)
k	Thermal conductivity	B	Boundary conditions differential operator
L	Thickness (m)	Greek letter	
τ	Non-dimensional time	θ	Neural network parameters
M_0	Initial moisture content (kg/kg)	λ	Loss weights
T_0	Initial temperature (K)	Ω	Computational domain
ΔT	Temperature difference	ϵ	Transport coupling coefficient
$PINN$	Physics-Informed Neural Network	δ	Thermogradient coefficient $\left(\frac{^\circ M}{K}\right)$
		α	Thermal diffusion
		$\mathcal{D}$	Differential operator

demonstrated the effectiveness of AutoML in optimizing heat and mass transfer processes within adsorption beds, achieving remarkable accuracy with models like Light Gradient Boosted Trees Regressor. Similarly, Krzywanski et al. [18] explored the potential of AutoML in enhancing the performance of adsorption cooling and desalination systems, showcasing its broad applicability and ability to handle complex optimization problems. Among these innovations, PINNs [19] have emerged as powerful tools for solving complex physical problems, accurately predicting and simulating a wide range of scenarios. This paradigm shift in computational science holds profound implications for a wide array of disciplines, with particular significance in the domain of life science engineering [20–22]. PINNs represent a groundbreaking advancement in the realm of surrogate modeling, seamlessly blending deep learning capabilities with the governing physical laws of heat and mass transfer. By integrating the principles of conservation of mass, energy, and momentum directly into the neural network architecture, PINNs harness the power of data-driven modeling while ensuring adherence to the underlying physics of the system. Some scholars focused on the application of PINNs to estimate the solution of the heat and mass transfer problems where a moderate amount of labeled data is used in training models [23,24]. The application of PINNs was explored in the simulation of heat transfer in porous media [25] and inverse heat transfer problems [26,27]. Pang et al. [28] used PINN in reconstructing of temperature field and identifying thermophysical parameters. On the other hand, other scholars were more interested in solving forward problems without any labeled data, such as in our previous study where we solved heat transfer equations in 1D, 2D, and 3D case studies by applying the boundary conditions in a hard manner [29].

Parameterized surrogate modeling is another application of deep learning techniques [30], which primarily aims at design and control problems. It relies on extensive, repeated simulations with the underlying parameterized numerical or experimental model over a large range of parameter values to discover the wide variety of parameter-dependent dynamical behaviors. However, within the PINNs framework, developing a data-free parameterized surrogate model is straightforward. This is achieved by incorporating the design parameter into the model's input and solving the governing equations for a limited number of design parameters. As a result, PINNs have been used in a variety of applications, such as uncertainty quantification in fluid dynamics [31]. This involves assessing the impact of variations in viscosity and geometric parameters on the fluid dynamics outcomes, such as velocity field and minimum wall shear stress. In heat transfer problems,

parameterized PINNs was used to solve heat transfer in a composite material for a wide range of varying heat transfer coefficients [32]. Further investigation in this field has been conducted by Würth [33] by employing PINNs in the optimization of a thermochemical manufacturing process with fiber-reinforced composite materials.

In continuation to our previous study [29], which focused on solving a single heat transfer equation, this paper investigates a novel application of PINNs for parameterized surrogate modeling of coupled and multiscale heat and mass transfer problems specifically in the context of drying processes. Drying processes are critical in various industries, including food processing, pharmaceuticals, and materials engineering, where precise control over moisture content and temperature is essential for product quality and energy efficiency. The application of PINNs in drying processes is particularly promising because it allows for the integration of physical laws directly into the neural network architecture, ensuring more efficient and cost-effective predictions. Our focus is on developing a surrogate model that can quickly predict temperature and moisture content over a specified range of Biot numbers for heat (Bi_t) and mass transfer (Bi_m) without extensive computational costs. This approach addresses some of the limitations of conventional numerical methods, such as high computational costs and difficulties in solving inverse problems. By incorporating parameters such as Bi_t and Bi_m , our PINNs model can simulate the drying process under various conditions, enabling rapid and reliable predictions without requiring extensive labeled data.

Previous studies have applied PINNs to heat and mass transfer problems, but there is a gap in the literature regarding their use in drying processes. In this study our aim is to fill this gap by demonstrating the efficacy of PINNs in providing fast and cost-efficient predictions of moisture content and temperature distribution during drying. We also compare our results with those obtained from conventional numerical methods and analytical solutions, highlighting the advantages of our approach in terms of computational efficiency.

Physical model and methodology:

The physical model involves a porous sample exposed to convective heat and mass transfer. We assume that the length of the sample is significantly larger than its thickness, with both sides exposed to convective heat and mass transfer. Consequently, we solve the governing equations for half of the sample and consider symmetry in the middle of the sample.

1.1. Governing equations

The governing equations are the coupled heat and mass transfer for one-dimensional unsteady problems. These equations are based on a modified version of the Luikov model, as presented in [34].

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + \frac{Lv}{C_p} \epsilon \frac{\partial M}{\partial t} \quad (1)$$

$$\frac{\partial M}{\partial t} = D \frac{\partial^2 M}{\partial x^2} + D \delta \frac{\partial^2 T}{\partial x^2} \quad (2)$$

Since the deep learning optimization methods work most effectively on dimensionless data, we derive the dimensionless relations here. The governing equations are transferred into a dimensionless form using the following dimensionless variables:

$$X = \frac{x}{L}, T^* = \frac{T - T_0}{T_a - T_0} = \frac{T - T_0}{\Delta T} \text{ and } M^* = \frac{M}{M_0} \quad (3)$$

$$\tau = \frac{\alpha t}{L^2}, Bi_t = \frac{hL}{K} \text{ and } Bi_m = \frac{h_m L}{D} \quad (4)$$

The Bi_t and Bi_m represent the thermal Biot number and mass transfer Biot number, respectively. The Luikov number and two other additional numbers, namely, Kossovitch number (Ko) and Posnov number (Pn) appear in this model, defined as:

$$Lu = \frac{D}{\alpha}, Ko = \frac{LvM_0}{C_p \Delta T} \text{ and } Pn = \frac{\delta \Delta T}{M_0} \quad (5)$$

By substituting the above non-dimensional variables into the original equations, the following correlations are derived:

$$\frac{\partial T^*}{\partial \tau} = \frac{\partial^2 T^*}{\partial X^2} + \epsilon Ko \frac{\partial M^*}{\partial \tau} \quad (6)$$

$$\frac{\partial M^*}{\partial \tau} = Lu \frac{\partial^2 M^*}{\partial X^2} + Lu Pn \frac{\partial^2 T^*}{\partial X^2} \quad (7)$$

The boundary conditions and initial conditions are expressed as:

$$at X = 1 : \frac{\partial T^*}{\partial X} = Bi_t (T_a^* - T^*), \quad (8)$$

$$\frac{\partial M^*}{\partial X} = Bi_m (M_a^* - M^*) - Pn \frac{\partial T^*}{\partial X}$$

$$at X = 0 : \frac{\partial T^*}{\partial X} = 0 \text{ and } \frac{\partial M^*}{\partial X} = 0 \quad (9)$$

1.2. Physics-informed neural network

In the context of physics-informed neural networks (PINNs), the solutions of coupled heat and mass equations are approximated with deep neural networks. The loss functions include all governing equations and physical constraints. PINNs consist of an input layer, one or more hidden layers, and an output layer. The input layer represents the independent variables of the problem, such as time, spatial coordinates, or any other desired variable involved in the physical constraints. The output layer represents the dependent variables, such as temperature or concentration. The key idea behind PINNs is to train the neural network to approximate the solution to the problem without needing labeled data by enforcing the governing equations as constraints during the training process. The mathematical formulation of PINNs can be represented as follows:

Let $T^*(X, \tau)$ be the solution to the PDEs we are interested in. This set of PDEs can be expressed with a differential operator (D) applied to the unknown solution.

$$D\{T^*(X, \tau)\} = f(X, \tau), X \in \Omega, \tau \geq 0 \quad (10)$$

$$B\{T^*(X, \tau)\} = g(X, \tau), X \in \partial\Omega, \tau > 0 \quad (11)$$

$$T^*(X, 0) = T_0^*, \tau = 0 \quad (12)$$

Where $f(X, \tau)$ and $g(X, \tau)$ are source terms and B is the differential operator of the boundary conditions. We define a neural network $N_T(X, \tau; \theta)$ with parameter θ that approximates a solution $T^*(X, \tau)$ for the unknown solution $T^*(X, \tau)$.

$$T^*(X, \tau) \approx \widehat{T^*(X, \tau)} = N_T(X, \tau; \theta) \quad (13)$$

The loss function of the deep neural network is defined as:

$$Loss(X, \tau; \theta) = \lambda_1 \|D\{T^*(X, \tau)\} - f(X, \tau)\| + \lambda_2 \|B\{T^*(X, \tau)\} - g(X, \tau)\| + \lambda_3 \|T^*(X, 0) - T_0^*\| \quad (14)$$

Each λ_i represents the weight assigned to individual terms that collectively construct the total loss function. The solution of this equation with the complementary boundary and initial conditions is found by minimizing the loss functions expressed as:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} Loss(X, \tau; \theta) \quad (15)$$

θ^* denotes the optimum values for the neural network's parameters, achieved when the loss reaches its minimum value. The most common method for training neural networks in general, including PINNs, involves utilizing the Adam optimizer [35], a first-order stochastic gradient descent technique. Due to its scalability, this approach has been successfully applied to many supervised learning tasks. In this study, we use the Adam optimizer with an initial learning rate of 10^{-3} to train our networks.

In the case of the drying process, the unknown solutions are $T^*(X, \tau)$ and $M^*(X, \tau)$. It is common practice in multi-objective forward problems to define one neural network with two outputs or two separate neural networks for each output variable with independent parameters. Here, we define two deep neural networks to estimate the solutions (Fig. 1.) as:

$$\widehat{T^*(X, \tau)} : (X, \tau) \mapsto N_T(X, \tau; \theta_1) \quad (16)$$

$$\widehat{M^*(X, \tau)} : (X, \tau) \mapsto N_M(X, \tau; \theta_2) \quad (17)$$

The loss function eventually is defined as:

$$Loss = \lambda_1 \left\| \frac{\partial T^*}{\partial \tau} - \frac{\partial^2 T^*}{\partial X^2} - \epsilon Ko \frac{\partial M^*}{\partial \tau} \right\| + \lambda_2 \left\| \frac{\partial M^*}{\partial \tau} - Lu \frac{\partial^2 M^*}{\partial X^2} - Lu Pn \frac{\partial^2 T^*}{\partial X^2} \right\| + \lambda_3 \left\| \frac{\partial T^*}{\partial X} - Bi_t (T_a^* - T^*) \right\| + \lambda_4 \left\| \frac{\partial M^*}{\partial X} - Bi_m (M_a^* - M^*) + Pn \frac{\partial T^*}{\partial X} \right\| + \lambda_5 \left\| \frac{\partial T^*}{\partial X} \right\| + \lambda_6 \left\| \frac{\partial M^*}{\partial X} \right\| \quad (18)$$

The PINNs implementation was carried out using SciANN [36] package. We used two separate deep neural networks for each output variable, with six hidden layers and 100 neurons for each layer. The activation functions are hyperbolic tangent with local adaptive weights. For the case of parameterized PINNs based on Bi_m and Bi_t , these parameters were seamlessly integrated into the model by being added individually to the input variables. This augmentation allowed our PINNs to explicitly account for the influence of Bi_t and Bi_m on the heat and mass transfer phenomena, thereby improving the accuracy and robustness of our predictions.

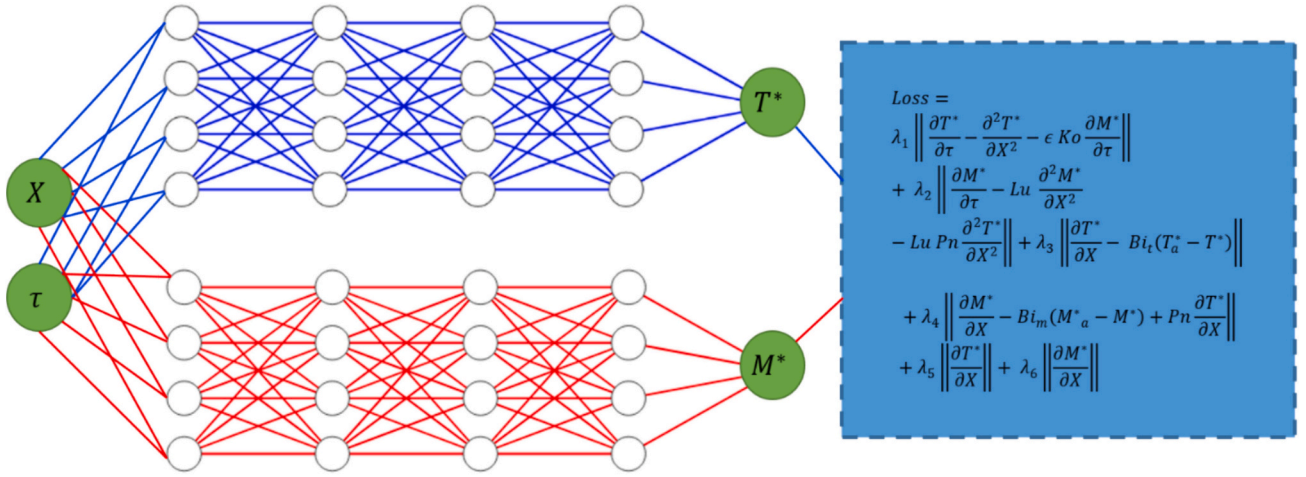


Fig. 1. PINNs architecture is used to solve heat and mass transfer equations with two independent neural networks.

2. Results and discussion

2.1. Validating models for convective heat transfer equations

To assess the accuracy of our model, we begin by validating its performance in solving a heat transfer problem involving forced convection. In this scenario, we consider a composite material exposed to forced convection with a heat transfer coefficient set to 5. The ambient temperature undergoes a gradual increase from 5 to 75 degrees Celsius. To establish the model's reliability, we compare its results with reference values obtained from the finite element method (FEM) [32]. The comparison, illustrated in Fig. 2, demonstrates a strong agreement between the model's predictions and the FEM reference values. This robust agreement validates the model's capability in accurately solving steady convective heat transfer problems.

2.2. Validating models for convective heat and mass transfer equations against experimental data

The accuracy of the model in solving coupled heat and mass transfer problems is assessed by comparing it with a reference study on dimensionless analysis of simultaneous heat and mass transfer in barley drying [37]. A transient mathematical model describing the isotropic and spherical grain drying process is formulated with governing equations and boundary conditions outlined as follows:

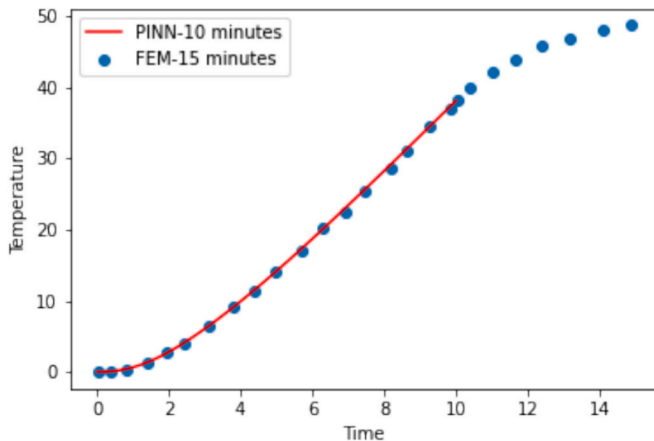


Fig. 2. Temperature evolution

$$\frac{\partial [C_p \rho T]}{\partial t} = k \left(\frac{\partial^2 T}{\partial r^2} + \frac{2}{r} \frac{\partial T}{\partial r} \right) + h_{fg} \times \rho \times \frac{\partial M}{\partial t} \quad (19)$$

$$\frac{\partial M}{\partial t} = D \left(\frac{\partial^2 M}{\partial r^2} + \frac{2}{r} \frac{\partial M}{\partial r} \right) \quad (20)$$

$$\text{atr} = r_0 : k \frac{\partial T}{\partial r} + h[T - T_\infty] + \rho h_{fg} \frac{V}{A} \frac{\partial M}{\partial t} = 0 \quad (21)$$

$$D \frac{\partial M}{\partial r} = h_m(M_e - M) \quad (22)$$

$$\text{atr} = 0 : \frac{\partial T}{\partial r} = 0,$$

$$\frac{\partial M}{\partial r} = 0$$

Where T and M represent the temperature and moisture content. V is the barley volume (m^3) and A is the barley area (m^2). h_{fg} denotes the latent heat of vaporization and h_m and h are the mass transfer coefficient and heat transfer coefficient, respectively. To solve the above equations with PINNs, all equations were converted in non-dimensional form, following a similar approach as the Luikov model. A comparison between moisture transportation on the mango surface obtained with PINNs and experimental data [38] is presented. The experiment by Sokhansanj and Bruce [38] involved spreading a single layer of grain, with initial moisture content ranging from 0.20 to 0.40 (dry basis), over a suspended wire mesh. It was then exposed to air at 0.6 m/s and temperatures between 40 and 175 °C. Weight was measured using an electronic balance, and surface temperature was recorded with an infrared pyrometer at 5-s intervals for 5 min, then every 30 s for 30 min, and finally at 1-min intervals. Airflow was stopped during weight measurements. Fig. 3 depicts the comparison of average moisture contents and temperature rise on the barley surface during drying. The relative error calculated between PINNs predictions and experimental data are 24% and 23% for moisture content and temperature, respectively.

2.3. Validating models using Luikov model and analytical solution

Given the relatively large errors observed in the initial comparison with experimental data, we continued our study by comparing the model's accuracy against analytical results [34]. This further comparison aims to validate the robustness and reliability of our PINNs model. The Luikov equations, widely used for describing complex heat and mass transfer phenomena in drying processes, serve as the basis for this

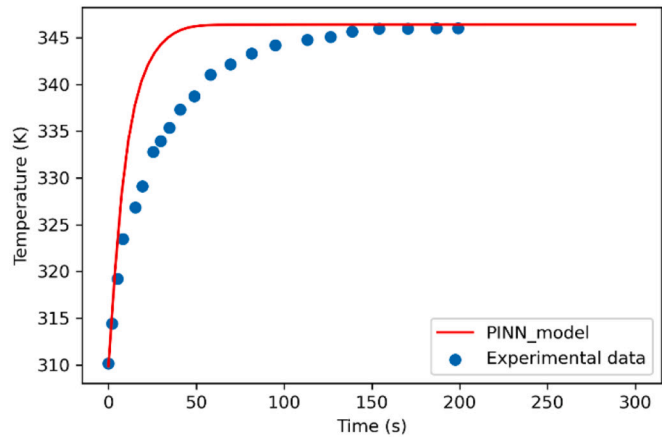
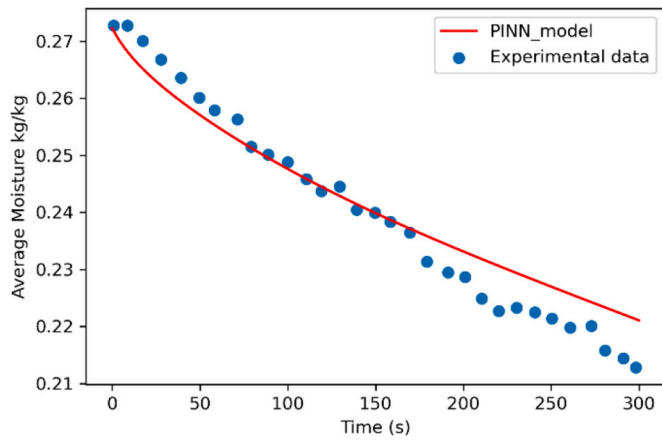


Fig. 3. comparison between average moisture contents in a barley seed.

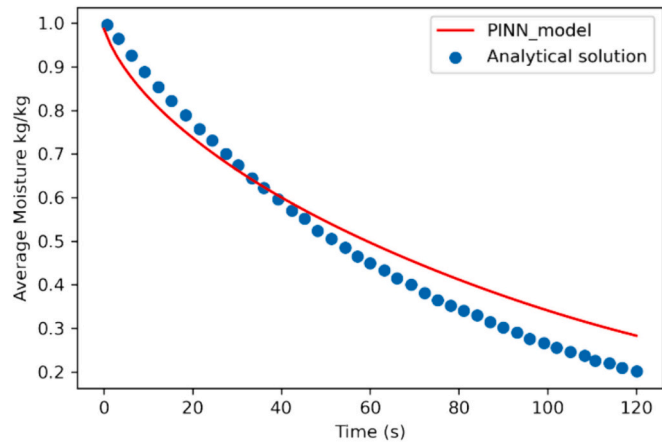


Fig. 4. Average moisture loss during drying predicted by PINNs model and analytical solution.

comparison. Figs. 4 and 5 display the average moisture loss and average temperature rise of olive pomace predicted by the PINNs model and analytical solution, respectively. The non-dimensional numbers in this study are $Ko = 10$, $Pn = 2$, $Lu = 0.005$, $Bi_t = 0.5$, and $Bi_m = 10$. These values were chosen based on their significant impact on drying processes, as highlighted in the reference study by Kokouch and et al. [34]. Specifically, the Luikov number (Lu) was set to 0.005 because lower Lu values have been shown to accelerate the drying process, which is crucial for accurately simulating the rapid drying kinetics. The

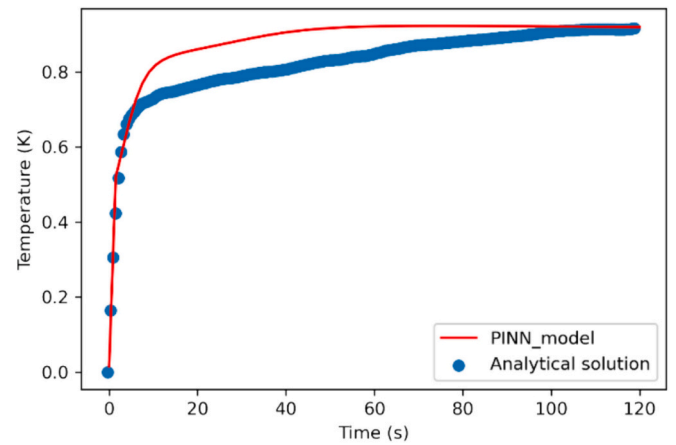


Fig. 5. Average temperature rise predicted by PINNs model and analytical solution.

Kossovitch (Ko) and Posnov (Pn) numbers were selected as 10 and 2, respectively, based on findings that specific combinations of Lu and Ko or Lu and Pn result in notable changes in the dimensionless average water content relative to the dimensionless average temperature, especially at the start of the drying process. The thermal Biot number (Bi_t) and mass transfer Biot number (Bi_m) were chosen as 0.5 and 10, respectively, to reflect realistic heat and mass transfer conditions typically seen in similar drying scenarios. These comparisons between analytical values and PINNs estimations verify the accuracy and reliability of the PINNs model in simulating multiscale heat and mass transfer phenomena in drying processes. The relative error between the PINNs predictions and reference values are 9.9% and 15% for moisture content and temperature, respectively.

In this section, a parameterized PINNs is developed to predict outcomes across a wide range of operational conditions characterized by varying Bi_m , while the Bi_t is constant 0.5. We utilized data from just four distinct Bi_t for training the model, selected as representative samples. Leveraging the parameterized PINNs framework, our model can forecast both temperature and moisture content for any input value of the Bi_m within the specified range (0 to 10), without incurring additional computational overhead (Fig. 6 and 7). Upon examination of the results, it was observed that at a Bi_m of zero, indicating negligible mass transfer, the average moisture content remained constant at a value of 1, while heat transfer persisted at a high rate. Furthermore, as the Bi_m , indicative of increased mass transfer, a corresponding augmentation in moisture transfer was observed, aligning with theoretical predictions. Interestingly, our analysis revealed that alterations in the Bi_m from 2.5 to 10 did

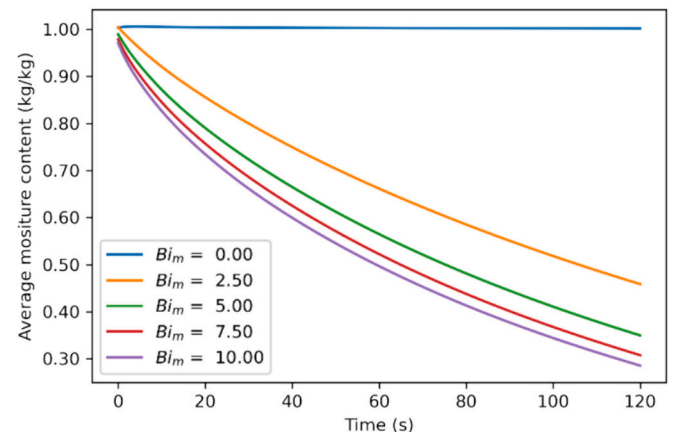


Fig. 6. Average moisture content variation at $Bi_t = 0.5$.

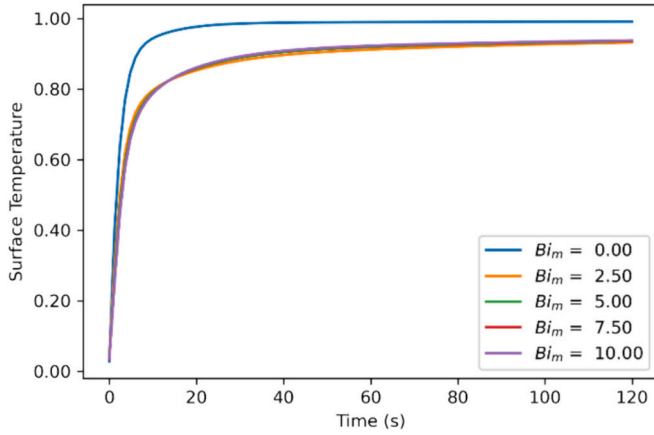


Fig. 7. Surface temperature variation at $Bi_t = 0.5$.

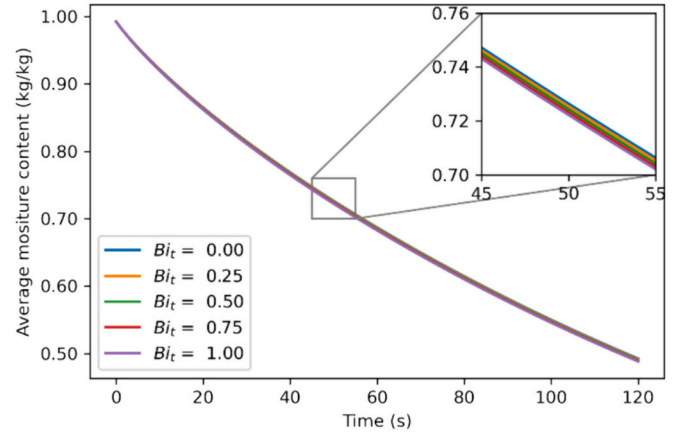


Fig. 9. Average moisture content variation at $Bi_m = 2$ by changing Bi_t .

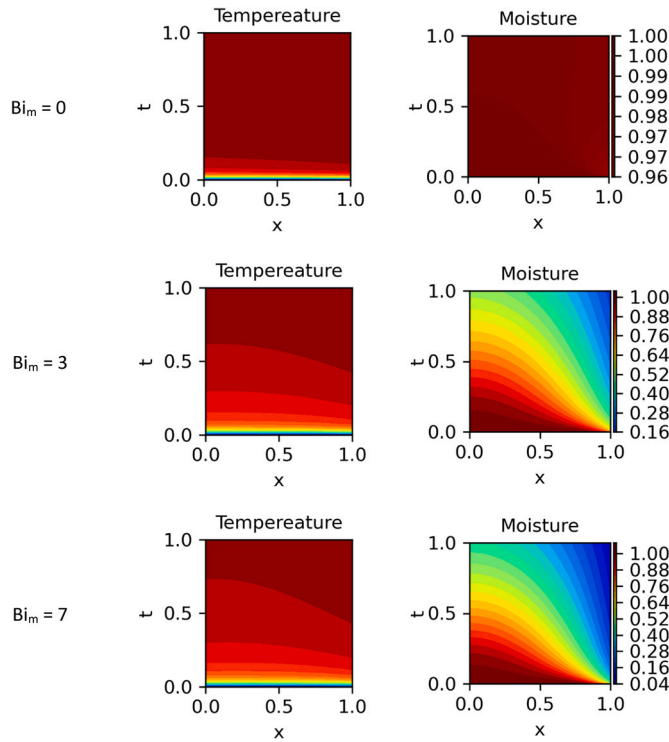


Fig. 8. Moisture and temperature distribution at $Bi_t = 0.5$ for Bi_m 0, 3 and 7.

not significantly affect the rate of heat transfer.

A comprehensive analysis of the moisture and temperature profiles is presented in contour form in Fig. 8, showcasing varying Bi_m of 0, 3, and 7, while keeping the Bi_t constant at 0.5. These profiles offer valuable insights into the progression of the drying process and can be instrumental in optimizing efficiency and quality. Our investigation reveals that the temperature profile exhibits a predominantly flat shape, indicating a robust heat conduction process. As distance from the heat source increases, there is a gradual decrease in temperature. Conversely, moisture content gradually changes as it diffuses within the material. Initially, moisture evaporates rapidly near the heat source, but this rate diminishes as we move away from the heat source. Yet, moisture diffusion through the material introduces a temporal aspect, causing the moisture content to decrease gradually over both distance and time. Consequently, the moisture profile in the x - t contour plot exhibits a curved trajectory, reflecting the interplay of moisture removal and diffusion processes.

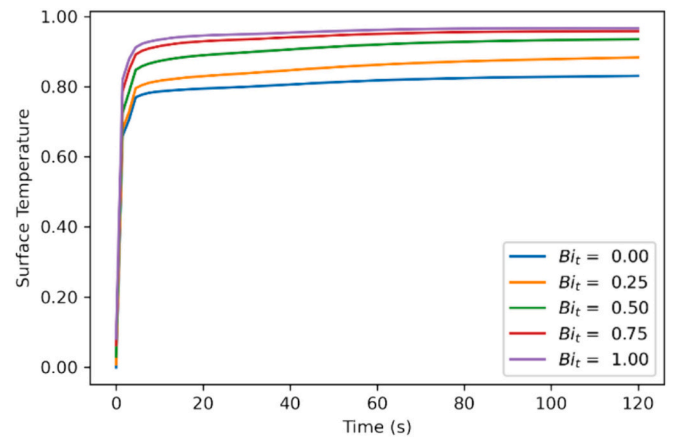


Fig. 10. Surface temperature variation at $Bi_m = 2$ by changing Bi_t .

A PINNs model, parameterized based on Bi_t , location (x) and time (t) is employed to estimate the temperature and the moisture content at varying Bi_t . Figs. 9, 10 summarize the surface temperature and average moisture content variation for a constant Bi_m of 2, while Bi_t ranges from 0 to 1. Our findings indicate that the moisture profiles exhibit minimal change with alterations in Bi_t changes with only a slight increase in moisture loss anticipated with higher Bi_t values. However, an increase in Bi_t leads to a rise in temperature, particularly evident at the onset of the process, characterized by a sharper temperature gradient.

In this study, we investigated the sensitivity of temperature and moisture content to variations in the Bi_t in a coupled heat and mass transfer system. To quantify this sensitivity, we calculated the derivatives of temperature and moisture content with respect to the Bi_t and plotted the results. We observed a distinct trend for the derivative of temperature with respect to the Bi_t . As the Bi_t varied from 0 to 1, the temperature derivative exhibited a parabolic curve, reaching a maximum value around $Bi_t = 0.3$. This peak indicates an optimal sensitivity range, where small changes in Bi_t impact temperature predictions most. Beyond this range, the sensitivity gradually decreases, suggesting diminishing effects on temperature distribution. In contrast, the derivative of moisture content with respect to the Bi_t exhibited a different behavior. Throughout the Bi_t range, the moisture content derivative remained nearly constant, around zero. This indicates that variations in the Bi_t had minimal impact on moisture content predictions in our study. The constant value suggests that changes in Bi_t did not significantly influence moisture distribution within the material under investigation. This finding helps us to gain a deep insight into optimum working conditions for the drying process without repeating the entire

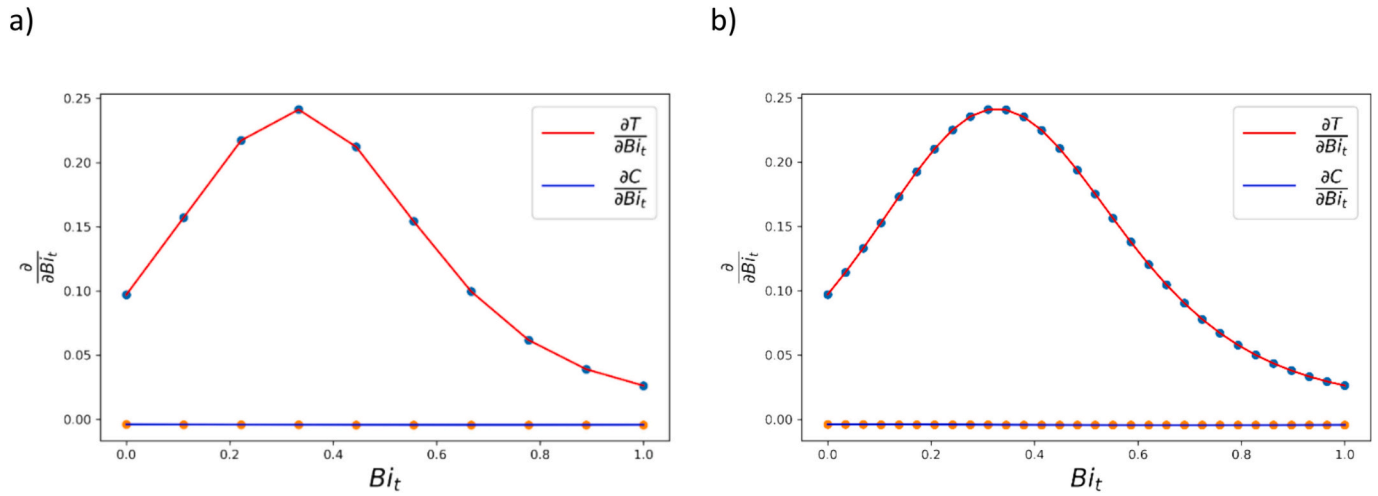


Fig. 11. Sensitivity of temperature and moisture with respect to Bi_t at the $x = 0.5$ and $t = 0.5$.

experiment or simulation for each single Bi_t . The initial PINNs model was trained with 5 Bi_t samples and the results of predictions on 10 and 30 Bi_t samples, respectively, are shown in Fig. 11a and 11b.

The sensitivity analysis was further extended to the variation of temperature and moisture content at a specific location and time, similar to a previous study. The Bi_m parameter was varied from 0 to 10, while Bi_t

was kept constant at 0.5. Initially, the PINNs model was trained with five samples of Bi_m , and the predictions were made on 15 Bi_m values, as shown in Fig. 12.a. The results indicated that temperature and moisture sensitivity at high Bi_m values approached a constant value near zero. However, both graphs had peaks and fluctuations within the range of 0 to 2. Increasing the number of samples to 30 made these fluctuations

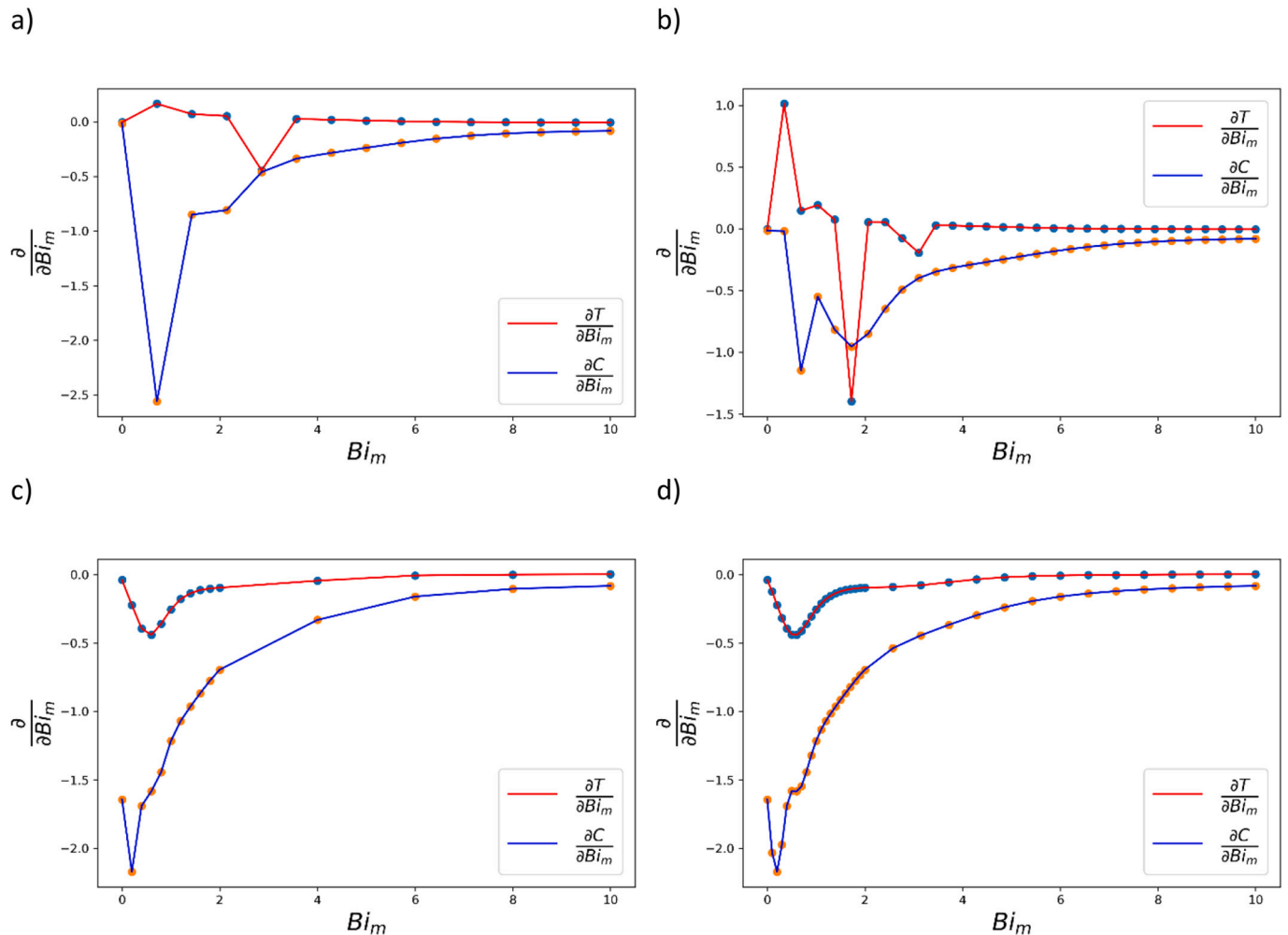


Fig. 12. Sensitivity of temperature and moisture with respect to Bi_m at $Bi_t = 0.5$.

more visible and sporadic, with no interpretable behavior (Fig. 12.b). This behavior mainly pointed out that the distribution of samples within this specific range was not fine enough to capture the dynamics of the problem. As a solution, a sample refinement was used to train the model. Using 10 samples specifically for this range (0,2) and 5 samples for the rest, a total of 15 samples were used for training the model. The results, as shown in Fig. 12.c, demonstrated a meaningful behavior in both graphs. The maximum sensitivity was observed in the vicinity of Bi_m zero, while this value in moisture content was notably larger than that in temperature, indicating that moisture content is much more sensitive to the variation of Bi_m . Moreover, by increasing the number of samples to 30 for predictions, we observed that both graphs became smoother (Fig. 12d), which proves that the sample refinement has improved the accuracy of predictions.

The topic of surrogate modeling, especially parameterized physics-informed machine learning, is vast and holds immense potential for various industrial sectors. Therefore, continuous study to discover new applications is an active area of research. Our future work will focus on the surrogate modeling of the drying process where the shrinkage of materials is also considered. A multiscale and multiphysics approach will be employed to further leverage the capabilities of PINNs. This will involve integrating additional physical phenomena, such as material deformation and phase changes, to enhance the model's accuracy and applicability. We aim to develop more sophisticated models that can handle complex boundary conditions, such as variable heat and mass fluxes, non-uniform drying environments, and dynamic changes in the drying medium. For example, in industrial drying processes, materials may experience fluctuating temperatures and humidity levels, leading to non-linear and time-dependent boundary conditions. By incorporating these complexities, we can provide more accurate and real-time predictions, thereby advancing the efficiency and effectiveness of industrial drying processes.

3. Conclusion

In conclusion, this study is the first to apply PINNs for surrogate modeling of the drying process without labeled data. By effectively integrating key parameters such as Bi_t and Bi_m within the model's architecture, we demonstrated the model's ability to predict temperature and moisture content with significant accuracy. The sensitivity analysis revealed temperature sensitivity peaks around Bi_t 0.3 and highlighted significant sensitivity to Bi_m in the early range of 0 to 0.5. These findings underscore the potential of the PINNs model to optimize drying processes and explore critical parameter sensitivities.

CRediT authorship contribution statement

Seyedalborz Manavi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ehsan Fattahi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Thomas Becker:** Writing – review & editing, Writing – original draft, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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